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Measurement of the azimuthal anisotropy of charged particles in $\sqrt{s_{\mathrm{NN}}} = 5.36$ TeV ^{16}O+^{16}O and ^{20}Ne+^{20}Ne collisions with the ATLAS detector

The ATLAS Collaboration

This paper presents the first measurements of the azimuthal anisotropy coefficients v_n, which quantify the n^{th}-order Fourier modulation of charged-particle azimuthal distributions, for $n = 2$–4 in $\sqrt{s_{\mathrm{NN}}} = 5.36$ TeV ^{16}O + ^{16}O and ^{20}Ne + ^{20}Ne collisions recorded with the ATLAS detector at the Large Hadron Collider in 2025. The v_n coefficients are measured as a function of transverse momentum (p_{T}), collision centrality, and event multiplicity. They are extracted using two complementary methods: two-particle correlations with a template-fit subtraction of short-range non-flow contributions, and four-particle subevent cumulants, which intrinsically suppress non-flow effects and provide sensitivity to flow fluctuations. The results show a clear hierarchy $v_2 > v_3 > v_4$ and a non-monotonic dependence on p_{T}, reaching a maximum around 2 GeV, consistent with trends observed in heavy-ion collisions. Detailed comparisons between the two collision systems reveal an enhanced v_2 in central ^{20}Ne + ^{20}Ne collisions, consistent with theory expectations based on the predicted prolate deformation of neon nuclei, in contrast to the slightly tetrahedral structure predicted for oxygen. The four-particle cumulant results highlight strong event-by-event fluctuations and provide the greatest sensitivity to nuclear shape effects. These measurements can place new constraints on the initial geometry and the hydrodynamic response in light-ion collisions, offering valuable input for models of nuclear structure.

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1 Introduction

A hot and dense state of nuclear matter in which the relevant degrees of freedom are strongly coupled quarks and gluons is known as the quark–gluon plasma (QGP). This state can be created transiently in collisions of heavy nuclei at high-energy colliders, such as the Relativistic Heavy Ion Collider (RHIC) and the Large Hadron Collider (LHC) [1–11]. The QGP produced in heavy-ion collisions undergoes collective expansion driven by strong pressure gradients, which convert initial-state spatial anisotropies into momentum anisotropies of the final-state hadron distribution [1–3, 12]. This phenomenon, commonly referred to as collective flow, is well described by nearly inviscid relativistic hydrodynamics and constitutes a key signature of QGP formation (see Ref. [13] and references therein).

The anisotropy of particle distributions in heavy-ion collisions is quantitatively characterized by a Fourier series in the azimuthal angle ϕ [14]:

$$\frac{dN}{d\phi} \propto 1 + 2\sum_{n=1}^{\infty} v_n \cos\left(n(\phi - \Psi_n)\right), \tag{1}$$

where v_n and Ψ_n represent the magnitude and orientation of the n^{th}-order anisotropy, respectively. The v_n are commonly referred to as "flow harmonics," while the Ψ_n are referred to as "event-plane angles." The v_n depend on transverse momentum p_{T}, pseudorapidity[1] (η), and event multiplicity, and fluctuate event-by-event (EbE) [15–18]. These EbE fluctuations arise primarily from variations in the initial geometry and energy density of the nuclear overlap region, which are driven by the fluctuating positions of nucleons and by subnucleonic structure, collectively referred to as geometric fluctuations. Among these coefficients, elliptic flow (v_2) is the largest due to the lenticular geometry of the average overlap region. This is typically followed by triangular flow (v_3), which typically has no contribution from the average geometry and is therefore generated entirely by EbE geometric fluctuations. For higher orders ($n \geq 4$), the flow harmonics are influenced not only by the initial geometry but also by non-linear mode coupling from lower-order harmonics, such as v_4 from v_2^2 and v_5 from $v_2 v_3$ [19]. These contributions make the higher-order harmonics sensitive to both the geometry and the medium's collective response. Extensive studies of v_n and their EbE fluctuations have provided important constraints on the initial-state geometry and on transport properties of the QGP in nuclear collisions, such as the shear viscosity to entropy density ratio η/s (see Ref. [13] and references therein).

The observation of collective flow was initially regarded as an exclusive signature of QGP formation in collisions of heavy nuclei, such as Au+Au or Pb+Pb systems. However, experiments at the LHC and the RHIC have revealed large v_n signals in much smaller collision systems, including pp and p+Pb at the LHC [20–23] and $p/d/^3$He+Au at the RHIC [24–27]. Remarkably, the observed v_n hierarchy in these small systems follows patterns consistent with expectations based on geometrical differences between their initial conditions. Detailed theory studies suggest that the measured v_n in small systems can be explained by the collective expansion of the matter produced in the collision, driven by the shape and fluctuations of the initial overlap region [28, 29]. Nevertheless, substantial uncertainties remain regarding the precise nature and properties of the produced matter in small collision systems. These uncertainties arise primarily from the incomplete understanding of the initial conditions in small systems, which are sensitive not only to the

[1] ATLAS uses a right-handed coordinate system with its origin at the nominal interaction point (IP) in the center of the detector and the z-axis along the beam pipe. The x-axis points from the IP to the center of the LHC ring, and the y-axis points upwards. Cylindrical coordinates (r, ϕ) are used in the transverse plane, ϕ being the azimuthal angle around the z-axis. The pseudorapidity is defined in terms of the polar angle θ as $\eta = -\ln\tan(\theta/2)$.

spatial distribution of nucleons within nuclei but also to the internal structure of individual nucleons [24, 30]. Consequently, precise characterization of the initial conditions in small systems is essential to improve the extraction of medium properties such as shear viscosity, pre-hydrodynamic evolution effects, and initial-state momentum anisotropies [31–33].

To address these uncertainties, a promising experimental strategy is to compare v_n harmonics in collisions of nuclei with similar mass numbers but distinct nucleon arrangements [34–36]. Such systems are expected to produce matter with comparable bulk properties and similar final-state collective responses. Therefore, the ratios of v_n between the two systems are expected to be primarily sensitive to differences between their initial conditions [37]. Analogous comparisons have previously been used to study the influence of nuclear structure in large collision systems [38–40]. A key advantage of applying this approach in small collision systems is that the initial-state differences are governed by nucleon distributions that can, in principle, be calculated using state-of-the-art *ab initio* nuclear structure theories [41], providing a theoretical foundation for interpreting experimental observations.

An ideal pair of collision systems for such comparative studies is ^{16}O+^{16}O and ^{20}Ne+^{20}Ne. Low-energy experiments and theory calculations indicate that ^{16}O has a near-spherical tetrahedral shape, whereas ^{20}Ne is predicted to consist of an ^{4}He cluster "orbiting" an ^{16}O core [42, 43]. Hydrodynamic simulations incorporating nuclear configurations from *ab initio* calculations predict significant enhancements in v_2 and v_3 for central ^{20}Ne+^{20}Ne collisions compared with central ^{16}O+^{16}O collisions[2] [44]. Furthermore, nuclear structure effects are also predicted to modify the EbE fluctuations of v_2 and v_3 between the two systems [45], providing additional discriminatory power for theory models.

This paper presents measurements of anisotropic flow coefficients v_n for $n = 2$ to 4 in $\sqrt{s_{\mathrm{NN}}} = 5.36$ TeV O+O and Ne+Ne collisions recorded by the ATLAS detector at the LHC in 2025. The analysis employs both two- and four-particle correlation methods [22, 46], which probe different moments of the v_n distributions. These measurements provide valuable constraints on the final-state collective response and on the properties of the produced medium in each system individually, while their comparison offers unique insights into the initial-state geometry and nucleon distributions within these light nuclei.

2 Experimental configuration

The ATLAS detector [47, 48] at the LHC covers nearly the entire solid angle around the collision point. It consists of an inner tracking detector surrounded by a thin superconducting solenoid, electromagnetic and hadronic calorimeters, and a muon spectrometer incorporating three large superconducting air-core toroidal magnets. The primary subsystems relevant to this study include the inner detector (ID), the calorimeter and the trigger and data acquisition infrastructure.

The ID is immersed in a 2 T axial magnetic field and provides charged-particle tracking in the range $|\eta| < 2.5$. The high-granularity silicon pixel detector covers the vertex region and typically provides four measurements per track, the first hit generally being in the insertable B-layer (IBL). It is followed by the SemiConductor Tracker (SCT), which usually provides eight measurements per track. These silicon detectors are complemented by the transition radiation tracker (TRT), which enables radially extended track reconstruction up to $|\eta| = 2.0$.

[2] For simplicity, ^{16}O and ^{20}Ne are hereafter denoted as O and Ne, respectively.

The calorimetry system includes several components: a liquid argon (LAr) electromagnetic calorimeter covering $|\eta| < 3.2$; a steel–scintillator sampling hadronic calorimeter covering $|\eta| < 1.7$; an additional LAr-based hadronic calorimeter for the region $1.5 < |\eta| < 3.2$; and forward LAr calorimeters (FCal) designed to measure both electromagnetic and hadronic activity in the range $3.2 < |\eta| < 4.9$. Forward neutrons produced from the breakup of nuclei in both hadronic and electromagnetic interactions are measured by compact tungsten sampling Zero Degree Calorimeters (ZDCs) positioned at $z = \pm 140$ meters from the ATLAS interaction point.

Event collection is handled by a two-tiered trigger system [49]. The first level (L1) is implemented through a combination of custom hardware and programmable logic, while the high-level trigger (HLT) uses software algorithms to refine the selection using more detailed detector information.

A software suite [50] is used in data simulation, in the reconstruction and analysis of real and simulated data, in detector operations, and in the trigger and data acquisition systems of the experiment.

3 Datasets, event and track selection

The O+O and Ne+Ne data used in this paper were collected in July 2025. The datasets correspond to integrated luminosities of 2 nb^{-1} for O+O and 0.5 nb^{-1} for Ne+Ne collisions, respectively. Minimum-bias events were selected with an L1 trigger based on the TRT "FastOR" algorithm [51], that required at least one TRT azimuthal sector above threshold, together with an HLT requirement of at least one reconstructed charged-particle track. To increase statistics for the highest-activity events, an additional trigger was used that required the TRT FastOR at L1 and at least 290 reconstructed tracks with $p_{\text{T}} > 200$ MeV at the HLT. A similar trigger with a requirement of 100 reconstructed tracks with $p_{\text{T}} > 200$ MeV at the HLT was employed to supplement the intermediate-activity region. In addition, a trigger requiring an L1 calorimeter transverse energy threshold of 20 GeV, followed by at least one reconstructed track with $p_{\text{T}} > 200$ MeV at the HLT, was also included.

In the offline analysis, the z-position of the primary vertex [52] was required to lie within 10 cm of the nominal interaction point (i.e. within 10 cm of the center of ATLAS). In the recorded O+O and Ne+Ne data, there are sizable contributions from pileup events, in which two or more inelastic collisions occur in the same bunch crossing. The majority of the pileup events are removed by requiring only a single high-quality reconstructed vertex per event. A high-quality vertex is defined as one that is well-constrained in position, having a z-position variance of less than 0.02 mm^2; such vertices have many associated tracks pointing to a common origin. Additional background is suppressed by removing events with significantly smaller track multiplicities than expected relative to the total transverse energy recorded in the FCal ($\Sigma E_{\text{T}}^{\text{FCal}}$) [53]. Similarly, correlations between the energy deposited in the ZDCs and $\Sigma E_{\text{T}}^{\text{FCal}}$ are used to suppress additional pileup by rejecting events for which the energy deposited in the ZDCs is significantly higher than in the majority of collisions. The estimated residual pileup after these selections is found to be at most 0.2%.

As in earlier ATLAS studies of heavy-ion collisions, events are categorized into centrality percentiles based on the $\Sigma E_{\text{T}}^{\text{FCal}}$ [5, 7, 15]. To relate the $\Sigma E_{\text{T}}^{\text{FCal}}$ distribution to the sampled fraction of the total inelastic O+O and Ne+Ne cross-sections, a Glauber-model–based calculation [54, 55] is used to fit the data, to extract the fraction of events selected above a minimum $\Sigma E_{\text{T}}^{\text{FCal}}$ threshold, and to estimate the systematic uncertainties on that fraction [5, 7, 15]. Additionally, the Glauber model is used to extract primary collision characteristics, such as the average number of participating nucleons, $\langle N_{\text{part}} \rangle$, for each centrality interval.

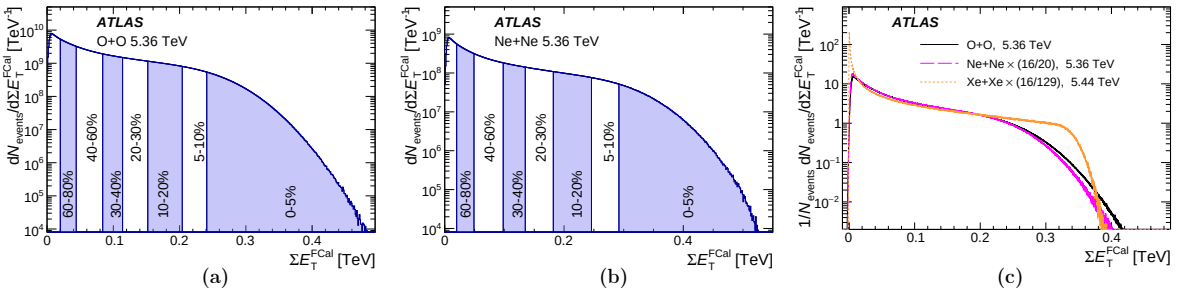

Figure 1: The $\Sigma E_{\mathrm{T}}^{\mathrm{FCal}}$ distribution in minimum-bias events, together with the thresholds for a few centrality intervals, for (a) O+O collisions and (b) Ne+Ne collisions. (c) Comparison to Xe+Xe collisions [7], where the $\Sigma E_{\mathrm{T}}^{\mathrm{FCal}}$ of the Xe+Xe and Ne+Ne systems are scaled by the relative number of nucleons to oxygen. The Ne+Ne and Xe+Xe distributions in panel (c) are normalized to have the same integral as the O+O distribution above 30 GeV.

The distribution of $\Sigma E_{\mathrm{T}}^{\mathrm{FCal}}$ observed in data, along with the threshold values defining various centrality intervals, is illustrated in Figure 1. Figure 1c shows the comparison of the $\Sigma E_{\mathrm{T}}^{\mathrm{FCal}}$ distributions in O+O and Ne+Ne, to that in Xe+Xe collisions (from Ref. [7]). The comparison is done after scaling the Ne+Ne and Xe+Xe $\Sigma E_{\mathrm{T}}^{\mathrm{FCal}}$ distributions by the number of nucleons relative to oxygen. While the Xe+Xe distribution exhibits a sharper fall-off at a scaled $\Sigma E_{\mathrm{T}}^{\mathrm{FCal}}$ of ~ 0.34 TeV and a narrower tail beyond this point, the O+O and Ne+Ne distributions show a significantly more pronounced high-$\Sigma E_{\mathrm{T}}^{\mathrm{FCal}}$ tail. This behavior reflects the larger relative event-by-event fluctuations in particle production in the smaller O+O and Ne+Ne systems compared to Xe+Xe.

Charged-particle tracks and collision vertices are reconstructed from hits in the ID using standard methods [56]. For the nominal analysis, the reconstructed tracks are required to have $p_{\mathrm{T}} > 0.5$ GeV, $|\eta| < 2.5$ and at least one pixel hit, with the additional requirement of a hit in the IBL when one is expected[3]. If a hit in the IBL is not expected, then a hit is required in the next-to-innermost pixel layer, if such a hit is expected. The tracks are required to have at least six SCT hits. To suppress secondary contributions, the transverse impact parameter of the track with respect to the beam-line, d_0, and the longitudinal impact parameter of the track relative to the primary vertex, $z_0 \sin(\theta)$, are required to satisfy $|d_0| < 1.5$ mm and $|z_0 \sin(\theta)| < 1.5$ mm. The quantity $N_{\mathrm{ch}}^{\mathrm{rec}}$ is defined as the number of charged-particle tracks in an event that satisfy these selection criteria and have $p_{\mathrm{T}} > 0.5$ GeV. An alternate set of more restrictive selections are used to evaluate systematic uncertainties in the measurement. For these "tight" selections, the number of pixel and SCT hits are raised to two and eight respectively, a requirement of at most one missing hit in the SCT is imposed, and the d_0 and $z_0 \sin(\theta)$ impact parameter selections are decreased to 1 mm. Furthermore, the χ^2 per degree of freedom of the reconstructed track trajectory is required to be less than 6.

Figures 2a and 2b show the correlation between $\Sigma E_{\mathrm{T}}^{\mathrm{FCal}}$ and $N_{\mathrm{ch}}^{\mathrm{rec}}$ in minimum-bias O+O and Ne+Ne events, respectively. These two measures of event activity are found to be well correlated. Figures 2c and 2d show the $N_{\mathrm{ch}}^{\mathrm{rec}}$ distributions in minimum-bias O+O and Ne+Ne events, together with the distributions for several centrality intervals from Fig. 1.

To study the detector performance, a sample of 7 million minimum-bias O+O Monte Carlo (MC) events was generated using the HIJING event generator version 1.38b [57]. Since HIJING does not have any intrinsic mechanism to generate flow, the latter is added after the initial particle generation step using an

[3] A hit is expected if the extrapolated track crosses an active region of a silicon-sensor module (pixel or SCT) that has not been disabled, and a hit is said to be "missing" when it is expected but not found. If a track crosses a disabled module, then for the purposes of hit counting, the disabled module is counted as a hit.

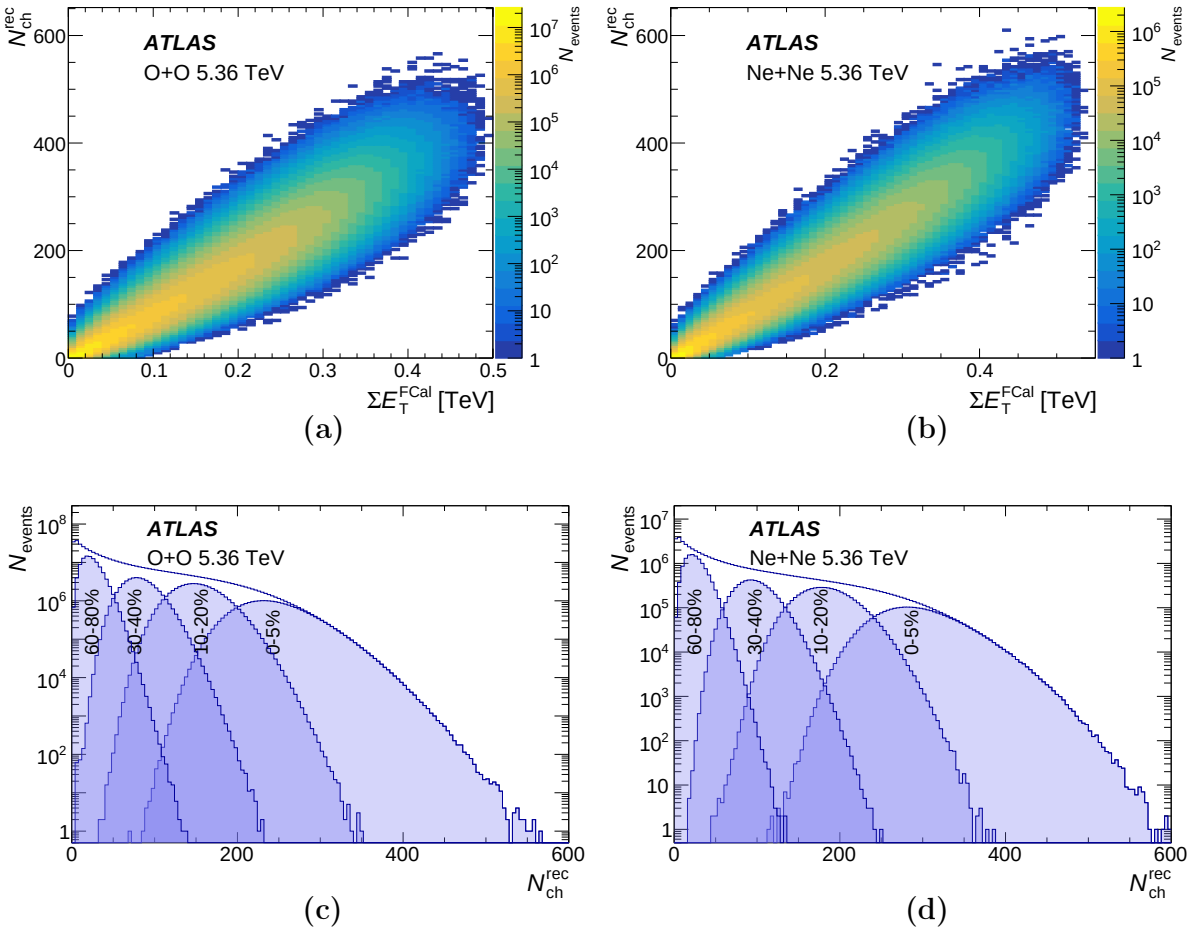

Figure 2: Correlation between N_{ch}^{rec} and ΣE_T^{FCal} in minimum-bias events for (a) O+O and (b) Ne+Ne collisions, and distributions of N_{ch}^{rec} in minimum-bias events for (c) O+O and (d) Ne+Ne collisions, including the distributions for several centrality intervals.

"afterburner" procedure [58], which slightly shifts the ϕ positions of generated particles to mimic flow. The generated sample was then passed through a full simulation of the ATLAS detector [59] using GEANT4 [60], and the MC events were reconstructed by the same algorithms as the data. The reconstructed particles in the MC events were used to calculate the reconstruction efficiency – the fraction of the generated charged particles that are successfully reconstructed and selected – as a function of p_T and η, and denoted by $\epsilon(p_T, \eta)$ below. With the criteria imposed in this analysis, the efficiency at $p_T = 1$ GeV varies between $\sim 65\%$ at $|\eta| = 2.5$ to $\sim 75\%$ at $|\eta| = 2$ and $\sim 85\%$ at $|\eta| = 0$. At mid-rapidity ($|\eta| < 1$) the efficiency is $\sim 80\%$ at $p_T = 0.5$ GeV and increases to $\sim 90\%$ at a p_T of 5 GeV. The rate of fake tracks, tracks that do not correspond to any generated particle, denoted by $f(p_T, \eta)$, is also estimated from the MC and stays below $\sim 2\%$ across the p_T and η ranges used in this measurement. The reconstructed event multiplicity N_{ch}^{rec} is corrected for reconstruction efficiency and fake tracks by weighting each track with:

$$\frac{1 - f(p_T, \eta)}{\epsilon(p_T, \eta)}, \tag{2}$$

where the numerator accounts for fake-track removal and the denominator for the reconstruction efficiency. This efficiency- and fake-corrected multiplicity is referred to as N_{ch}.

4 Methodology

Due to the limited multiplicity per event, the v_n cannot be reliably measured on an EbE basis. Instead, the flow harmonics v_n are estimated from multi-particle correlations, which average over many events and provide access to different moments of the EbE v_n distributions. However, correlations unrelated to collective flow can contaminate the measurement and must be suppressed or removed. For example, jet production and resonance decays produce strongly correlated collinear particles ($|\Delta\eta|, |\Delta\phi| < 1$), and weaker but equally important back-to-back correlations at $\Delta\phi \approx \pi$ that persist even at large $\Delta\eta$. These few-particle, non-global correlations are referred to as "non-flow." This section describes two approaches for estimating flow harmonics: the two-particle correlations (2PC) method, including its improved template-fit implementation, and the multi-particle cumulant method. In the 2PC approach, a template fit is used to subtract non-flow contributions based on the lowest-multiplicity events, while the multi-particle cumulant method applies a subevent technique to suppress non-flow effects.

4.1 Two-particle correlations and template fit

The 2PC method has been widely used for flow measurements at the RHIC and the LHC [6, 9, 15, 16, 21, 22, 61–69]. Correlations between pairs of charged particles are studied as a function of their relative pseudorapidity, $\Delta\eta = \eta^a - \eta^b$, and relative azimuthal angle, $\Delta\phi = \phi^a - \phi^b$. The indices a and b denote the two particles in the pair, whose kinematic selections may differ. To account for detector acceptance effects, the correlation function is defined as the ratio of the "same-event" pair distribution S, where both particles are taken from the same event, to the "mixed-event" distribution B, where they are taken from different events [15]:

$$C(\Delta\eta, \Delta\phi) = \frac{S(\Delta\eta, \Delta\phi)}{B(\Delta\eta, \Delta\phi)}.$$

The same-event distribution contains both genuine physical correlations and contributions from detector acceptance, inefficiencies and nonuniformities that are not related to the underlying physics. The mixed-event distribution reflects only these non-physical effects, so that their ratio isolates the genuine physical correlations [61]. To ensure this, events used for the B distribution are required to have similar centrality (or multiplicity) and vertex position. When constructing S and B, corrections for track reconstruction inefficiency and fake tracks are applied using a per-pair weight (see Eq. 2):

$$\frac{(1 - f(p_{\mathrm{T}}^a, \eta^a))(1 - f(p_{\mathrm{T}}^b, \eta^b))}{\epsilon(p_{\mathrm{T}}^a, \eta^a)\, \epsilon(p_{\mathrm{T}}^b, \eta^b)}.$$

Examples of $C(\Delta\eta, \Delta\phi)$ are shown in Figure 3, normalized such that the integral of $B(\Delta\eta, \Delta\phi)$ matches that of $S(\Delta\eta, \Delta\phi)$ for $|\Delta\eta| > 2$. In all cases a prominent peak is observed at $\Delta\eta = \Delta\phi = 0$, arising from short-range correlations such as jet fragmentation, resonance decays, or Hanbury Brown–Twiss (HBT) correlations [70]. At large $\Delta\eta$, long-range correlations are visible both on the near-side ($\Delta\phi \sim 0$) and the away-side ($\Delta\phi \sim \pi$). The near-side correlation, commonly referred to as the "ridge," originates primarily from collective flow. The away-side correlation receives contributions from both collective flow and back-to-back dijets.

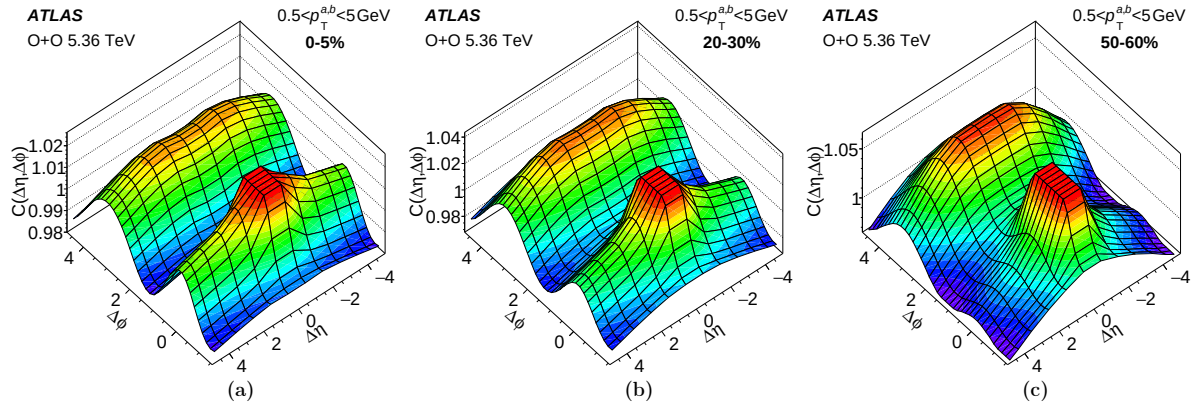

Figure 3: Two-particle $\Delta\eta - \Delta\phi$ correlations in O+O collisions for the (a) 0–5%, (b) 20–30%, and (c) 50–60% centrality intervals. The plots are for $0.5 < p_{\mathrm{T}}^{a,b} < 5$ GeV. The distributions are truncated along the z-axis, to suppress the peak at $\Delta\eta = \Delta\phi = 0$, and are plotted over $|\Delta\eta| < 4.5$ to avoid statistical fluctuations at larger $|\Delta\eta|$.

One-dimensional correlation functions, $C(\Delta\phi)$, are obtained by integrating the S and B distributions over the range $2 < |\Delta\eta| < 5$:

$$C(\Delta\phi) = \frac{\int_2^5 S(|\Delta\eta|, \Delta\phi)\, d|\Delta\eta|}{\int_2^5 B(|\Delta\eta|, \Delta\phi)\, d|\Delta\eta|} \equiv \frac{S(\Delta\phi)}{B(\Delta\phi)},$$

where the $|\Delta\eta| > 2$ requirement is imposed to suppress non-flow correlations arising from the peak at $\Delta\eta = \Delta\phi = 0$ seen in Figure 3 [21, 22, 62]. The $C(\Delta\phi)$ are normalized to have an average value of unity. Similar to the single-particle distribution (Eq. 1), the $C(\Delta\phi)$ is parametrized with a Fourier series [15]:

$$C(\Delta\phi) = C_0 \left(1 + 2\Sigma_{n=1}^{\infty} v_{n,n}(p_{\mathrm{T}}^a, p_{\mathrm{T}}^b) \cos(n\Delta\phi)\right). \tag{3}$$

To suppress residual non-flow contributions that persist over $|\Delta\eta| > 2$, primarily on the away side ($\Delta\phi \sim \pi$), a commonly used template-fit procedure is employed [22, 24, 64, 71]. In this method, the shape of the non-flow component is estimated from low-multiplicity (peripheral) events and assumed to be unchanged in higher-multiplicity (more central) events. For this analysis, the peripheral reference correlation, $C^{\mathrm{periph}}(\Delta\phi)$, is constructed from events of the same collision species (O+O or Ne+Ne) with centrality greater than 80%. The measured correlation $C(\Delta\phi)$ is then parameterized as the sum of this non-flow reference and an azimuthally modulated pedestal, $C^{\mathrm{ridge}}(\Delta\phi)$, that encodes the collective anisotropy:

$$C(\Delta\phi) = FC^{\mathrm{periph}}(\Delta\phi) + C^{\mathrm{ridge}}(\Delta\phi), \tag{4}$$

with

$$C^{\mathrm{ridge}}(\Delta\phi) \equiv G\left[1 + 2\sum_{n=2}^{5} v_{n,n}(p_{\mathrm{T}}^a, p_{\mathrm{T}}^b) \cos(n\Delta\phi)\right]. \tag{5}$$

The parameters F, G, and the $v_{n,n}$ are determined by the template fit, with F and G constrained such that the integrals of both sides of Eq. 4 are equal. Fourier terms up to fifth order ($v_{2,2}$-$v_{5,5}$) are included in the fit.

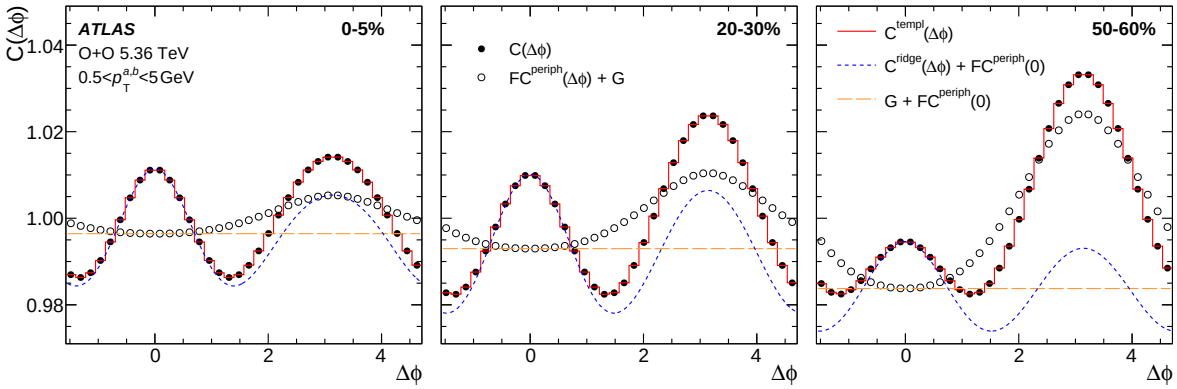

Figure 4: Template fits to correlation functions measured in O+O collisions for the (left) 0–5%, (center) 20–30%, and (right) 50–60% centrality intervals. The plots are for $0.5 < p_{\mathrm{T}}^{a,b} < 5$ GeV. The solid points indicate the measured $C(\Delta\phi)$, and the continuous red line indicates the template fit, $C^{\mathrm{templ}}(\Delta\phi)$. The open points and dashed curves indicate the different components of the template-fit, which are shifted along the y-axis by G or by $FC^{\mathrm{periph}}(0)$, where necessary, for presentation.

Figure 4 shows examples of template fits for O+O collisions, where the template fit is denoted as $C^{\mathrm{templ}}(\Delta\phi)$. In the measured correlations, the away-side peak is the largest in the 50–60% centrality interval and decreases systematically in the 20–30% and 0–5% intervals. A significant fraction of the away-side correlation is described by the scaled peripheral reference ($FC^{\mathrm{periph}}(\Delta\phi)$ term in Eq. 4). The relative contribution of this term decreases monotonically toward mid-central and central events, indicating that non-flow correlations are most important in peripheral collisions. The $C^{\mathrm{ridge}}(\Delta\phi)$ component of the template fit is double-peaked in all intervals, reflecting the dominant contribution from the $v_{2,2}$ term in Eq. 5.

If the pair distribution is entirely determined by a global single-particle distribution, like in Eq. 1, the Fourier coefficients of the $C(\Delta\phi)$ (or $C^{\mathrm{ridge}}(\Delta\phi)$) factorize into the product of single-particle anisotropies [61] as $v_{n,n}(p_{\mathrm{T}}^a, p_{\mathrm{T}}^b) = v_n(p_{\mathrm{T}}^a)v_n(p_{\mathrm{T}}^b)$ and thus:

$$v_n(p_{\mathrm{T}}^b) = \frac{v_{n,n}(p_{\mathrm{T}}^a, p_{\mathrm{T}}^b)}{v_n(p_{\mathrm{T}}^a)} = \frac{v_{n,n}(p_{\mathrm{T}}^a, p_{\mathrm{T}}^b)}{\sqrt{v_{n,n}(p_{\mathrm{T}}^a, p_{\mathrm{T}}^a)}}, \tag{6}$$

For all the 2PC results in this analysis, the $v_n(p_{\mathrm{T}}^b)$ are evaluated using Eq. 6 with $0.5 < p_{\mathrm{T}}^a < 5$ GeV. The upper limit on p_{T}^a is chosen to suppress non-flow, which increases at high p_{T}. The template-fit method provides a more reliable treatment of non-flow effects and is therefore regarded as the primary measurement method. Results obtained with the 2PC method are also included in this paper to illustrate the impact of non-flow removal achieved by the template-fit procedure.

4.2 Four-particle cumulants

Multi-particle cumulants extract higher-order azimuthal correlations, providing information about EbE flow fluctuations. The cumulant measurements have the advantage of suppressing correlations from jets and dijets, instead of relying on an explicit procedure to correct v_n as discussed in Section 4.1. The cumulant of order $2k$, where k is an integer, involves correlations between $2k$ particles and suppresses all correlations

involving less than $2k$ particles, including non-flow correlations [72]. The framework for the cumulant measurements is described in Refs. [73–75], but a concise description is provided here for completeness.

As mentioned before, the cumulant method involves the calculation of $2k$-particle azimuthal correlations $\langle\{2k\}_n\rangle$, and $2k$-particle cumulants, $c_n\{2k\}$, for the n^{th}-order flow harmonics, where k equals either one or two in this paper. The two- or four-particle azimuthal correlations in one event are evaluated as [73–75]:

$$\langle\{2\}_n\rangle = \left\langle e^{in(\phi_1-\phi_2)} \right\rangle \tag{7}$$

$$\langle\{4\}_n\rangle = \left\langle e^{in(\phi_1+\phi_2-\phi_3-\phi_4)} \right\rangle \tag{8}$$

where "$\langle\cdot\rangle$" denotes a single-event average over all pairs or quadruplets of distinct particles, respectively. The averages from Eqs. 7 and 8 are expanded into products of per-particle normalized flow vectors [76]:

$$\langle q_n\rangle = \left\langle e^{in(\phi)} \right\rangle, \tag{9}$$

which provides an efficient calculation of multi-particle correlations. The exact details of this procedure follow Ref. [46]. These flow vectors are constructed with per-particle weights that correct for detector non-uniformities, tracking inefficiency, and contributions from fake tracks, similar to Eq. 2 but with additional ϕ-dependent corrections applied.

In the "standard" cumulant method described so far, all $2k$-particle multiplets involved in the calculations of $\langle\{2k\}_n\rangle$ are selected using the entire detector acceptance. To further suppress the non-flow correlations, which typically involve particles emitted within a localized region in η, the particles can be grouped into several "subevents," each covering a non-overlapping η interval [75]. The multi-particle correlations are then constructed by correlating particles between different subevents (two and three in this case), further reducing non-flow correlations. For two-subevent correlations, particles 1 and 2 in Eq. 7 are selected from different regions of η. Similarly, in Eq. 8, the pair of particles 1 and 2 are selected from one region of η, and the pair of particles 3 and 4 from another. For the three-subevent correlations, permutations of different choices are made and combined (see Ref. [46] for details). For the results presented here, the subevents used for the two-subevent method cover $-2.5 < \eta < 0$ and $0 < \eta < 2.5$, and those for the three-subevent method cover $-2.5 < \eta < -2.5/3$, $-2.5/3 < \eta < 2.5/3$ and $2.5/3 < \eta < 2.5$.

The two- and four-particle cumulants are then obtained from the azimuthal correlations as:

$$c_n\{2\} = \langle\langle\{2\}_n\rangle\rangle, \tag{10}$$

$$c_n\{4\} = \langle\langle\{4\}_n\rangle\rangle - 2\langle\langle\{2\}_n\rangle\rangle^2, \tag{11}$$

where "$\langle\langle\cdot\rangle\rangle$" represents the average of $\langle\{2k\}_n\rangle$ over an event ensemble. In the absence of non-flow correlations, $c_n\{2k\}$ reflects the moments of the distribution of the flow coefficient v_n:

$$\begin{aligned} c_n\{2\} &= \langle v_n^2\rangle, \\ c_n\{4\} &= \langle v_n^4\rangle - 2\langle v_n^2\rangle^2 . \end{aligned} \tag{12}$$

The v_n measured by the 2-particle cumulants is defined as $v_n\{2\} = \sqrt{c_n\{2\}}$. In the absence on non-flow effects, the v_n measured by the 2-particle cumulant is identical to that measured using the 2PC method. However, non-flow contributions lead to differences between the two. Henceforth, the v_n measured with the 2PC and template-fit, methods are denoted as $v_n^{\text{2PC}}\{2\}$ and $v_n^{\text{sub}}\{2\}$, respectively. When making general

statements without reference to a specific method, v_n obtained from any two-particle correlation technique is denoted simply as $v_n\{2\}$.

Under the Gaussian model of eccentricity fluctuations [76], the 2- and 4-particle cumulants can be expressed in terms of an average geometry-driven component $\bar{x}_n$ and a fluctuation component Δ_n as:

$$c_n\{4\} = -\bar{x}_n^4, \quad c_n\{2\} \equiv \langle v_n^2 \rangle = \bar{x}_n^2 + \Delta_n^2. \tag{13}$$

This implies $c_n\{4\} < 0$, and a positive $c_n\{4\}$ signals onset of non-Gaussian flow fluctuations, or significant non-flow contamination. If the sign constraints are obeyed, the corresponding four-particle flow coefficient is then defined as:

$$v_n\{4\} = \sqrt[4]{-c_n\{4\}}, \tag{14}$$

and measures the contribution to the flow from the average-geometry component only.

A comparison of the results of the standard and the two- and three-subevent cumulants in the O+O and Ne+Ne collisions shows that the two- and three-subevent results are consistent. This demonstrates that both subevent methods effectively suppress non-flow in the measured phase space region. For this reason, the $v_n\{4\}$ results presented in this paper uses the two-subevent method.

5 Systematic uncertainties

The systematic uncertainties of the measured v_n using the 2PC, template-fit, and multi-particle cumulant methods are described in this section. The following sources of systematic uncertainty are considered:

1. **MC closure:** The MC closure test compares v_n^{gen}, obtained from MC generated particles, with v_n^{reco}, obtained by applying the full analysis procedure to reconstructed tracks in the MC simulation, as done in data. The difference between the two is within 1% across the p_{T} and multiplicity ranges considered in this analysis and is conservatively assigned as a systematic uncertainty. This uncertainty accounts for residual reconstruction effects not corrected in the data analysis.

2. **Track selection:** The track selection criteria control the relative contributions of genuine charged particles and fake tracks entering the analysis. The stability of the results with respect to the track selections is evaluated by varying the requirements applied to reconstructed tracks and including the resulting variation in v_n as a systematic uncertainty. The results obtained with the nominal selections are compared with those using the tighter criteria described in Section 2. The differences depend on the harmonic order and are between 0.5–1.5%.

3. **Tracking efficiency:** The uncertainty in the reconstruction efficiency and fake-rate due to ID material modeling in the GEANT4 simulation is accounted for by evaluating the efficiency and fake-rates in alternate MC samples. In each sample, a single modification is applied to the ATLAS ID geometry: the passive material of the ID is increased by 5%, or the passive material of the IBL by 10%, or the passive material in the services region by 25%. These variations capture the full range of data–MC differences observed in dedicated studies of the ID material [77]. The variation in the results when using these alternative models is taken as a systematic uncertainty. This uncertainty is less than 0.25%.

Table 1: The contributions to the systematic uncertainty of v_n in O+O collisions from different sources, as function of centrality. The contributions are expressed in percentages. Items 1–5 are common to all the methods used here (2PC, template-fit and cumulants). Item 6 is specific to the 2PC and template-fit methods. Item 7 is specific to the template-fit method. Item 8 is specific to the cumulant method. The uncertainties are shown for the integrated p_{T} interval of 0.5–5 GeV.

Source	harmonic order	0–40% [%]	40–70% [%]
1. MC closure	v_2–v_4	1	1
2. Track selection	v_2	0.5	0.5
	v_3	0.75	0.75
	v_4	1.5	1.5
3. Tracking efficiency	v_2–v_4	0.25	0.25
4. Centrality definition	v_2	0.2	0.2–0.6
	v_3	0.2–1.0	1–2
	v_4	0.2	0.2–0.6
5. Residual pileup	v_2–v_4	0.2	0.2
6. Event-mixing	v_2	0.25	0.25
	v_3	0.5	0.5
	v_4	1	1
7. Peripheral reference	v_2	0.5	0.5–3
	v_3	0.75–3.5	3.5–12
	v_4	1.0–4.5	4.5–20
8. Flattening procedure	v_2–v_3	0.25	0.25

4. **Centrality definition:** The centrality definitions used to classify the events into centrality percentiles have a ~1% (2%) uncertainty associated with them in the O+O (Ne+Ne) measurements. This arises from uncertainties in the fraction of the inelastic O+O and Ne+Ne cross-sections accepted by the triggers used in this analysis and is estimated from the Glauber fits to the $\Sigma E_{\mathrm{T}}^{\mathrm{FCal}}$ distributions [5, 6]. The impact of this uncertainty on the v_n is evaluated by varying the $\Sigma E_{\mathrm{T}}^{\mathrm{FCal}}$ thresholds that define the centrality intervals, re-evaluating the v_n, and assigning the observed variation as a systematic uncertainty. This uncertainty is negligible in most central collisions, and increases systematically for more peripheral collisions.

5. **Residual pileup:** Pileup events dilute the measured v_n as there are no correlations between independent collisions. The estimated residual pileup is at most 0.2% in any centrality or multiplicity interval considered in this paper. Because the maximum possible dilution cannot exceed the pileup rate, the entire residual pileup rate of 0.2% is conservatively assigned as a systematic uncertainty on the v_n.

6. **Event-mixing:** As mentioned before, the 2PC method uses event-mixing to account for detector acceptance effects. The nominal mixing matches events that are within a z_{vtx} separation of less than 20 mm. Alternate mixing criteria, where the matching is restricted to within 10 mm and relaxed to 200 mm are used, and the maximum variation in the results is included as a systematic uncertainty. This uncertainty is of order 1% and only affects the 2PC and template-fit measurements.

7. **Peripheral reference:** For the centrality-dependent measurements, the nominal template-fit procedure uses events more peripheral than 80% for building the peripheral reference $C^{\text{periph}}(\Delta\phi)$. For the multiplicity-dependent results, the peripheral references are built from events from the same collision system that have $N_{\text{ch}}^{\text{rec}}$ less than 20. For evaluating uncertainties associated with the assumptions made in the template fit analysis, the centrality and multiplicity dependence measurements are repeated with the peripheral reference built from 5.02 TeV pp events with $N_{\text{ch}}^{\text{rec}}$ less than 20, as was done in Ref. [7]. The difference between the results with this alternate choice of peripheral reference are included as a systematic uncertainty. This uncertainty is relevant only for the template-fit method.

8. **Flattening procedure:** The cumulant measurements use a flattening procedure to remove detector non-uniformities in ϕ [46]. As a conservative estimate of the systematic related to the flattening procedure, the measurements were repeated with the flattening removed, and the resulting variations of $\sim 0.25\%$ are included as a systematic uncertainty on the $v_n\{4\}$.

Table 1 summarizes the final systematic uncertainties for the integrated p_{T} interval of 0.5–5.0 GeV in the O+O measurements. Except for the uncertainties related to the peripheral reference and centrality definition, all other uncertainties are conservatively taken to be constant and sufficiently large to cover the variations across all centrality (or multiplicity) and p_{T} intervals studied here. The dominant uncertainties are the uncertainties related to the MC-closure and the peripheral reference variation. For Ne+Ne, the assigned uncertainties are taken to be the same as for O+O, with values chosen large enough to cover both systems, except for the centrality-definition uncertainty, which is about twice as large. These individual uncertainties are added in quadrature to obtain total uncertainties. For ratios of the Ne+Ne to O+O v_n measurements, all sources of systematic uncertainty are treated as correlated, with the exception of the residual pileup. The latter is not considered correlated, since pileup rates can differ between the two systems due to their distinct running conditions.

6 Results

Figure 5 shows the measured $v_n^{\text{sub}}\{2\}$ and the $v_n^{\text{2PC}}\{2\}$ as a function of p_{T}^b in O+O and Ne+Ne collisions at $\sqrt{s_{\text{NN}}} = 5.36$ TeV. Comparing the template fit v_n and the 2PC v_n, a large potential non-flow contribution is observed in the 2PC results, particularly at higher p_{T}^b and in peripheral collisions. The v_2 results exhibit a linear rise at low p_{T}^b, followed by a decrease in the range of 2–4 GeV for the 0–5% and 20–30% centrality intervals. This behavior is qualitatively similar to what is observed in other collision systems, from Pb+Pb to pp [6, 7, 22, 64]. A hierarchy is also observed, with the magnitude of v_n decreasing as n increases, in agreement with observations from other systems. In peripheral events (50–60% centrality), or at p_{T} above approximately 3 GeV, significant differences are observed between the 2PC and template-fit measurements, highlighting the impact of non-flow background correlations on the 2PC method.

The hydrodynamic response to the initial geometry is best studied as a function of variables that are directly impacted by it, chiefly charged-particle multiplicity and centrality. Figure 6a shows $v_n^{\text{sub}}\{2\}$ and $v_n\{4\}$ for charged particles with $0.5 < p_{\text{T}} < 5.0$ GeV in O+O and Ne+Ne collisions as a function of N_{ch}. In both systems, $v_n^{\text{sub}}\{2\}$ is observed to exceed $v_n\{4\}$ for the same harmonic order n, which reflects the positive contribution of fluctuations to $v_n\{2\}$ as described in Eq. 13. The difference between the $v_2\{2\}$ and $v_2\{4\}$ increases with increasing multiplicity indicating the increased role of the fluctuation component, Δ_2^2 in Eq. 13, relative to the mean geometry component, $\bar{x}_2^2$, going from peripheral to central collisions.

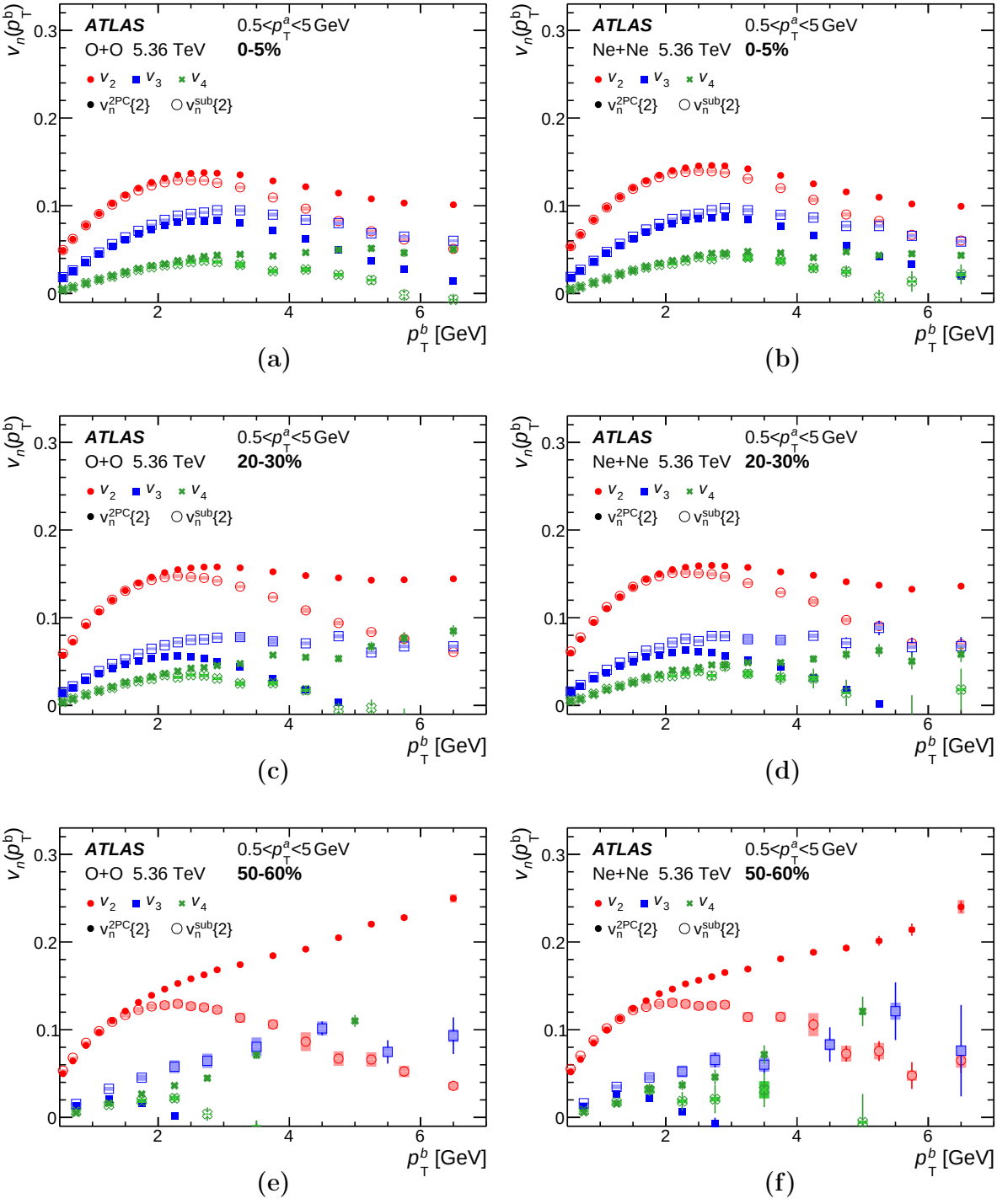

Figure 5: The p_{T}^{b} dependence of the $v_n^{\mathrm{2PC}}\{2\}$ and $v_n^{\mathrm{sub}}\{2\}$ in the (a, c, e) O+O and (b, d, f) Ne+Ne collisions, for the (a, b) 0–5%, (c, d) 20–30% and (e, f) 50–60% centrality intervals. The solid and open points show the results obtained using the 2PC and the template-fit methods, respectively. The vertical lines and vertical bars indicate statistical and systematic uncertainties, respectively.

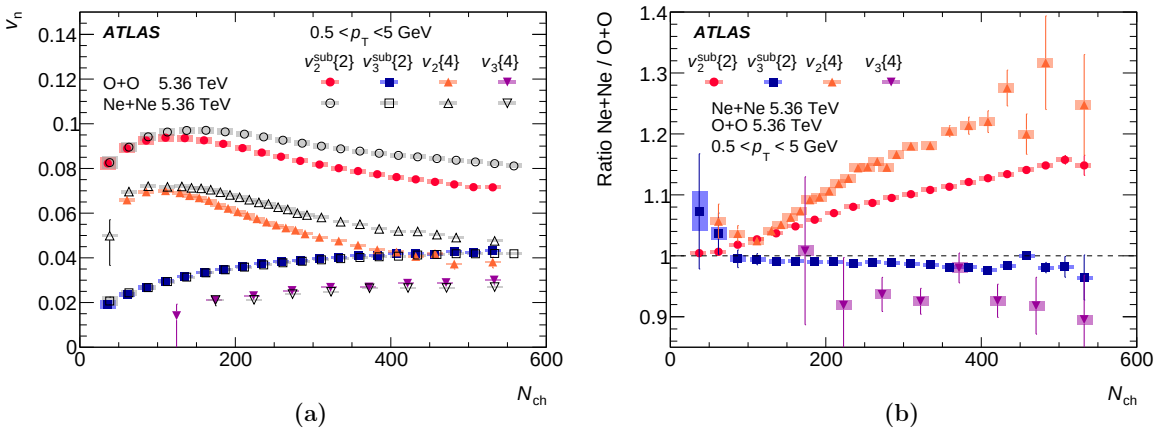

Figure 6: (a) Flow harmonics $v_n^{\mathrm{sub}}\{2\}$ and $v_n\{4\}$ for the elliptic and triangular moments in O+O collisions (colored solid markers) and Ne+Ne collisions (open black markers) at $\sqrt{s_{\mathrm{NN}}}$ = 5.36 TeV, shown as a function of N_{ch}. The Ne+Ne points are slightly displaced along the horizontal axis for visual clarity. (b) Ratios of the flow coefficients in Ne+Ne to those in O+O collisions, corresponding to the results in panel (a). Ratios are shown only for the 0–50% centrality range to suppress large statistical fluctuations in more peripheral intervals. The vertical lines and vertical bars indicate statistical and systematic uncertainties, respectively. For the ratios, systematic uncertainties correlated between the O+O and Ne+Ne measurements largely cancel.

To compare O+O and Ne+Ne v_n directly, their ratio is plotted in Figure 6b. For the elliptic flow, v_2, there is a growing enhancement in Ne+Ne collisions with increasing N_{ch}. These can be attributed to centrality-dependent changes in the overlap geometry and neon's deformed nuclear structure, which will be further discussed below, in the context of the centrality-dependent results. Conversely, there is a striking similarity of $v_3^{\mathrm{sub}}\{2\}$ in O+O and Ne+Ne collisions as a function of N_{ch}. This provides evidence that the hydrodynamic response and the fluctuation-driven component of the initial-state geometry depend primarily on the particle density. The residual difference and systematically smaller $v_3\{4\}$ may be an indication of differing average triangularity in the initial-state geometry.

To compare the geometric aspects of oxygen and neon nuclei, Figure 7a presents the results as a function of collision centrality, using the same kinematic requirements and analysis techniques as in Figure 6. Although light-ion collisions have large initial-state geometry fluctuations and possibly nuclear deformations, a large contribution to the elliptic geometry is the event-averaged lenticular shape. This event-averaged shape depends on the ratio of the impact parameter to the nuclear radius, which is highly correlated with centrality. For this reason there is closer agreement observed between O+O and Ne+Ne as a function of centrality than N_{ch}. Thus, the ratio of Ne+Ne to O+O flow harmonics is calculated as a function of centrality, in an attempt to remove these average geometric effects. This ratio is shown in Figure 7b. For both $v_2^{\mathrm{sub}}\{2\}$ and $v_2\{4\}$, a relatively centrality-independent ratio is observed for centralities more peripheral than 10%, as opposed to the ratios as a function of N_{ch} (Figure 6b). However, over the 0–10% centrality interval, a growing enhancement appears toward the most central collisions. This enhancement is more pronounced in the four-particle cumulant measurement. A similar behavior is observed for the triangular moment: the $v_3^{\mathrm{sub}}\{2\}$ ratio exhibits a sudden decrease in the 0–10% centrality interval. For the $v_3\{4\}$ ratio, the statistical uncertainties are too large to observe clear trends.

Figure 8 shows the comparisons of the measured $v_n^{\mathrm{sub}}\{2\}$ with the template-fit method to $v_n\{2\}$ calculations from the model described in Ref. [78]. The model combines the Projected Generator Coordinate Method (PGCM), an *ab initio* nuclear structure model for oxygen and neon, with the IPGlasma framework,

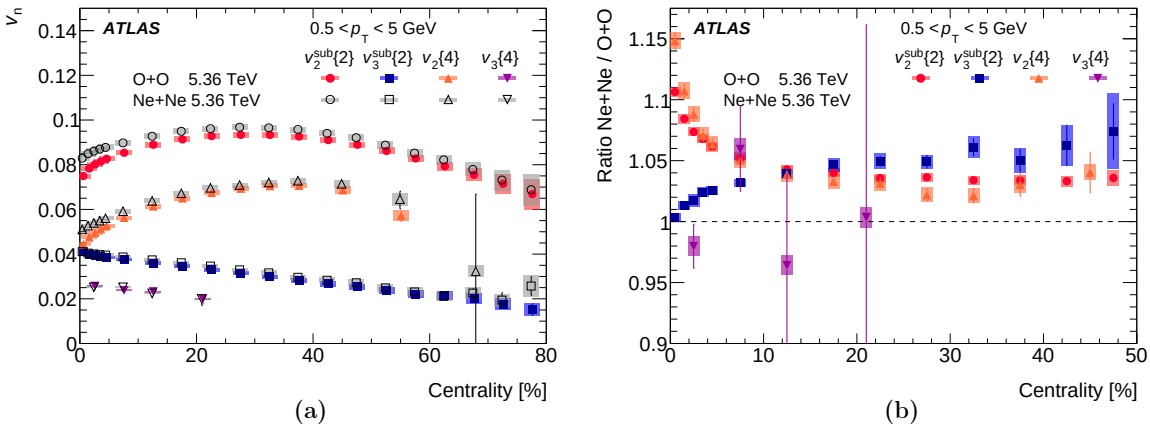

Figure 7: (a) Flow harmonics $v_n^{\mathrm{sub}}\{2\}$ and $v_n\{4\}$ for the elliptic and triangular moments in O+O collisions (colored solid markers) and Ne+Ne collisions (open black markers) at $\sqrt{s_{\mathrm{NN}}}$ = 5.36 TeV, shown as a function of centrality. The Ne+Ne points are slightly displaced along the horizontal axis for visual clarity. (b) Ratios of the flow coefficients in Ne+Ne to those in O+O collisions, corresponding to the results in panel (a). The vertical lines and vertical bars indicate statistical and systematic uncertainties, respectively. For the ratios, systematic uncertainties correlated between the O+O and Ne+Ne measurements largely cancel.

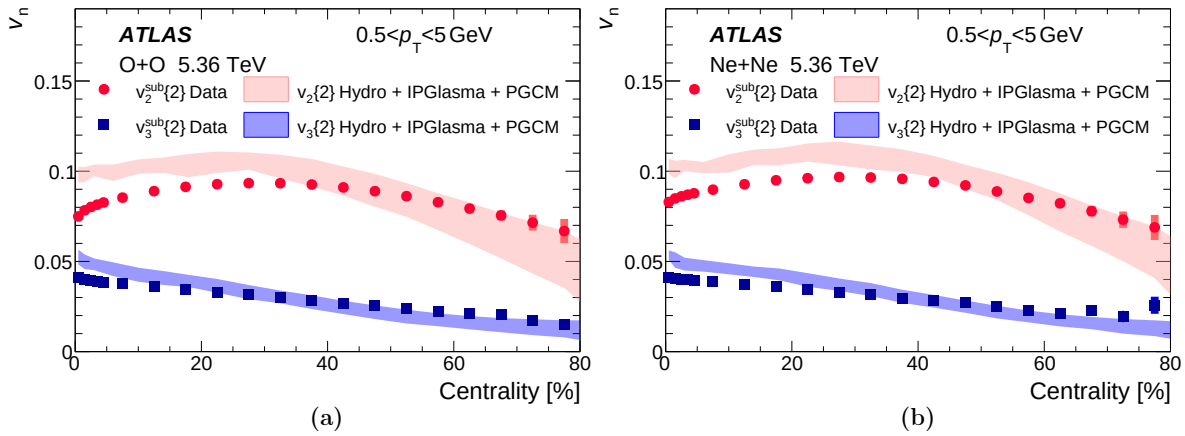

Figure 8: Comparison of $v_2^{\mathrm{sub}}\{2\}$ and $v_3^{\mathrm{sub}}\{2\}$ with theory calculations from the model described in Ref. [78], for (a) O+O and (b) Ne+Ne collisions. The theory calculations are labeled as "Hydro+IPGlasma+PGCM." For the data, the vertical lines and vertical bars indicate statistical and systematic uncertainties, respectively. The data uncertainties are sometimes too small to be visible. The bands for the theory calculations represent the combined statistical and systematic uncertainties.

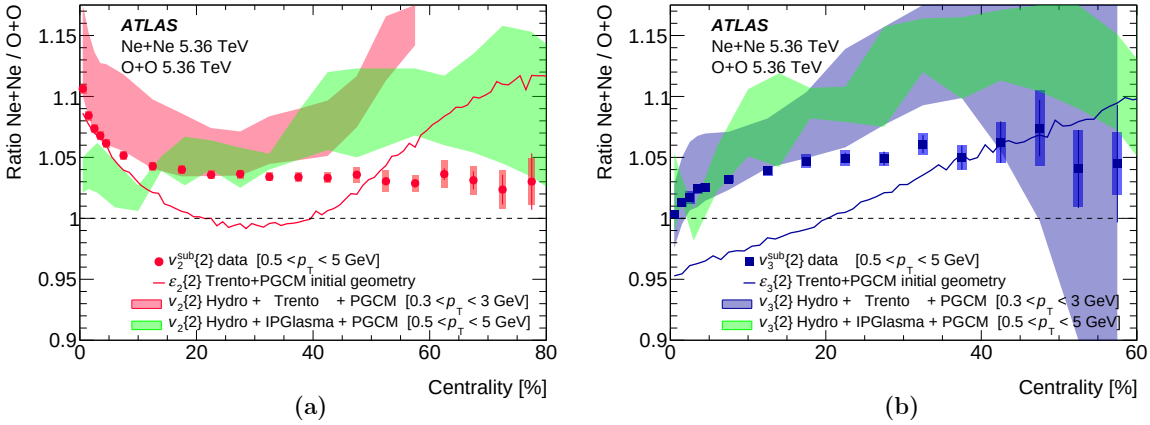

Figure 9: Comparison of the measurements to theory predictions for the ratio of (a) $v_2\{2\}$ and (b) $v_3\{2\}$ between Ne+Ne and O+O. The $v_n\{2\}$ in the data are obtained from the template fit method. The Hydro+Trento+PGCM theory is taken from Ref. [44] and the Hydro+IPGlasma+PGCM is an extension of Ref. [78]. The ratios are also shown for the eccentricities, a quantification of the elliptic (ε_2) or triangular (ε_3) shape of the initial-state energy density, using the Trento+PGCM model. For the data, the vertical lines and vertical bars indicate statistical and systematic uncertainties, respectively. The bands for the theory calculations represent combined statistical and systematic uncertainties.

including JIMWLK evolution [79], to calculate the initial energy densities of the colliding nuclei. These energy densities are then evolved using viscous relativistic hydrodynamics (MUSIC [80]). Finally, a hadronic afterburner (UrQMD [81, 82]) is applied to obtain the v_n. The centrality is determined with the charged-particle multiplicity at mid-rapidity. The model is denoted as "Hydro+IPGlasma+PGCM" in the figure. The calculations overpredict the v_2 and v_3 in central collisions, but are consistent within uncertainties with the measurements in more peripheral collisions. These comparisons indicate that the light-ion measurements can provide additional constraints for tuning existing theory models.

Figure 9 shows comparisons of the measured ratios for $v_n\{2\}$ between Ne+Ne and O+O collisions, to two theory calculations. The first is the Hydro+IPGlasma+PGCM model introduced above. The second model is described in Ref. [78] and uses PGCM [83] to determine the nuclear configurations of both oxygen and neon, which are then collided, generating initial-state energy densities with the Trento model [84]. These events are then simulated with the Trajectum hydrodynamic framework [85, 86]. This model is labeled as "Hydro+Trento+PGCM." The model uses mid-rapidity multiplicity to determine centrality. The fluctuations in the model calculations are statistical in nature. The measured $v_n\{2\}$ are obtained from the template-fit method. The event-averaged eccentricity, a quantification of the magnitude of ellipticity ϵ_2 and the triangularity ϵ_3, ratios for a particular Trento parameter set [87] are also shown in Figure 9. The data and Hydro+Trento+PGCM model agree quantitatively, at the edge of the theory uncertainties, in $v_2\{2\}$ in central collisions. The large enhancement in the $v_2\{2\}$ ratio in the 0–10% central collisions observed in the hydrodynamic theory calculation is attributed to the elongated shape of the neon nucleus, leading to an elliptic initial-state, as can be seen in the $\epsilon_2\{2\}$ ratio. The qualitative behavior of the measured $v_2\{2\}$ ratio supports this picture. The Hydro+IPGlasma+PGCM calculation does not capture this proposed geometrically driven behavior in the 0–10% centrality range. This inability to capture geometric effects in the latter calculation, is also reflected in the data-theory disagreement in Figure 8. In both Ne+Ne and O+O collisions, the measured v_2 decreases in the most central collisions, but this trend is not predicted by the model. Because the nuclear structure model PGCM can capture

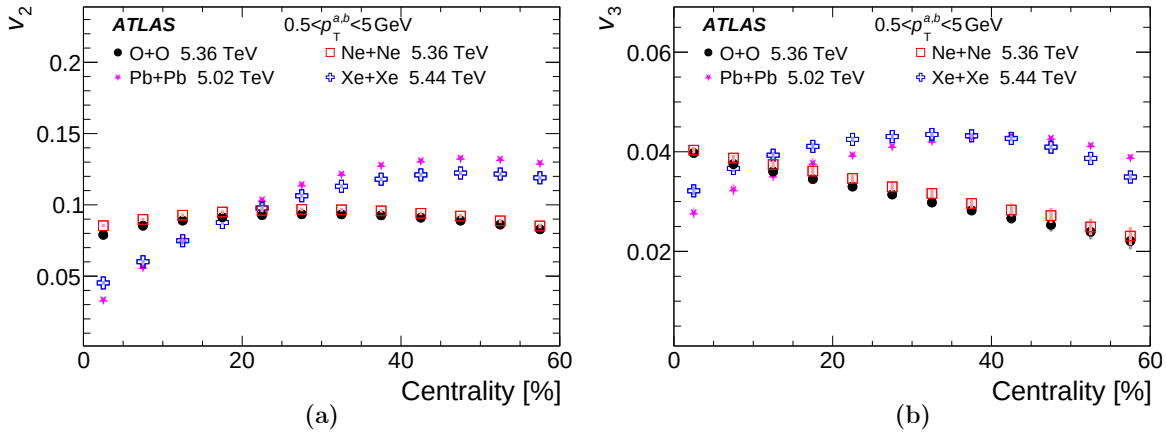

Figure 10: Comparisons of the (a) v_2 and (b) v_3 measured by the template-fit method in O+O and Ne+Ne collisions with prior measurements in Xe+Xe and Pb+Pb collisions from Ref. [7], as a function of centrality. The Xe+Xe and Pb+Pb results are obtained with the 2PC method. The plots are for $0.5 < p_{\mathrm{T}}^{a,b} < 5$ GeV. The vertical lines and vertical bars indicate statistical and systematic uncertainties, respectively.

geometric effects, as seen in the Hydro+Trento+PGCM v_2 ratio agreement with the data, the disagreement with the Hydro+IPGlasma+PGCM calculations could arise in the IPGlasma-with-JIMWLK portion of the initial-state model. It has been noted [78] that this aspect of the model creates large multiplicity fluctuations, reducing the correlation between the centrality and impact parameter. This can lead to the inability of the model to reproduce the centrality dependence trends observed in Figures 8 and 9. For both model comparisons presented in Figure 9, the experimental uncertainties are significantly smaller than the statistical and systematic uncertainties associated with the model calculations. Consequently, further refinement of the theory predictions is required to fully capitalize on the precision of the data.

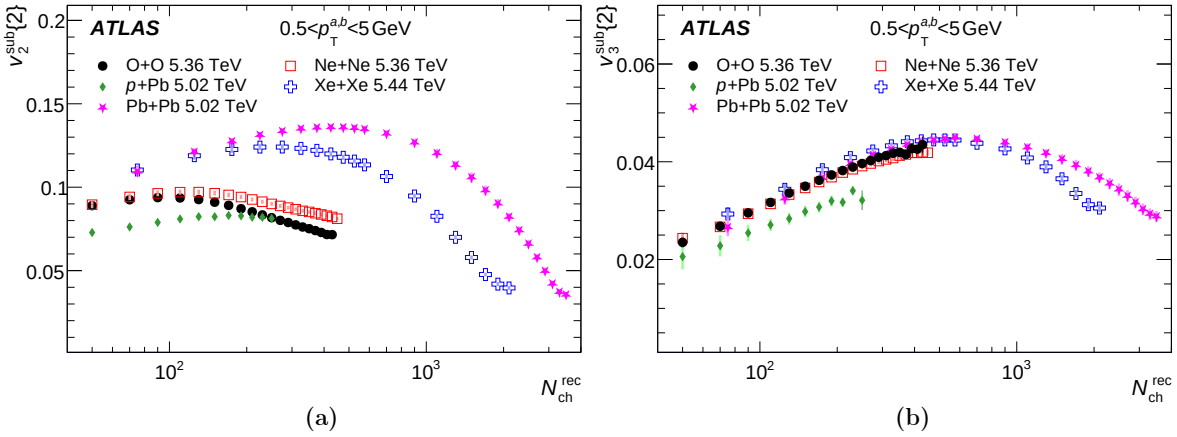

Figure 11: Comparison of the (a) $v_2^{\mathrm{sub}}\{2\}$ and the (b) $v_3^{\mathrm{sub}}\{2\}$ measured in O+O and Ne+Ne collisions with prior measurements in p+Pb, Xe+Xe and Pb+Pb collisions from Ref. [7], as a function of the reconstructed charged-particle multiplicity ($N_{\mathrm{ch}}^{\mathrm{rec}}$). The vertical lines and vertical bars indicate statistical and systematic uncertainties, respectively, and are often too small to be visible.

Figure 10 shows comparisons of $v_n^{\mathrm{sub}}\{2\}$ measurements in O+O and Ne+Ne collisions to previous $v_n^{\mathrm{2PC}}\{2\}$ results in Xe+Xe collisions at 5.44 TeV and Pb+Pb collisions at 5.02 TeV from Ref. [7], as a function of centrality. In heavy-ion collisions, the v_2 is largely determined by the elliptic geometry of the nuclear

overlap, which is strongly correlated with centrality and thus drives the pronounced centrality dependence of v_2. By contrast, the v_2 in light-ion systems shows a much weaker centrality dependence, highlighting the dominant role of EbE geometry fluctuations in these systems. The v_3 in Xe+Xe and Pb+Pb collisions exhibits a non-monotonic centrality dependence, increasing from central to mid-central collisions and then decreasing. In light-ion collisions, however, the v_3 decreases monotonically from central to peripheral events. Figure 11 shows similar comparisons of $v_n^{\mathrm{sub}}\{2\}$ as a function of reconstructed charged-particle multiplicity, $N_{\mathrm{ch}}^{\mathrm{rec}}$, including results from p+Pb collisions at 5.02 TeV. The most notable feature is the very similar magnitude and $N_{\mathrm{ch}}^{\mathrm{rec}}$ dependence of v_3 in both light- and heavy-ion collisions. This stands in stark contrast to the centrality dependence of v_3, which differs strongly between light- and heavy-ion systems. Interestingly, the p+Pb v_3 is also comparable in magnitude to the light- and heavy-ion results at the same multiplicities and shows a qualitatively similar multiplicity dependence.

7 Conclusion

This paper presents the first comprehensive measurements of anisotropic flow coefficients v_n (n = 2–4) in $\sqrt{s_{\mathrm{NN}}}$ = 5.36 TeV O+O and Ne+Ne collisions with the ATLAS detector at the LHC. The results are obtained using two-particle (template-fit) and four-particle (subevent cumulant) methods, and explore the dependence of v_n on transverse momentum, multiplicity, and centrality in each system. To isolate the role of initial-state geometry, ratios of Ne+Ne to O+O v_n are also studied.

The measurements reveal a characteristic rise and fall of v_2 with p_{T}, peaking at 2–4 GeV, large event-by-event fluctuations in the most central collisions, and a clear hierarchy $v_2 > v_3 > v_4$, consistent with hydrodynamic expectations. Multiplicity- and centrality-dependent ratios of $v_2\{2\}$ and $v_2\{4\}$, taken between Ne+Ne and O+O collisions, reveal a marked enhancement in Ne+Ne, consistent with the elongated nuclear shape of neon and reproduced by model calculations. For v_2, the four-particle cumulant ratios provide the greatest sensitivity to these geometric effects. For $v_3\{2\}$, the Ne+Ne values exceed those in O+O for most centralities, except in the 0–1% most central events where they are comparable. Detailed comparisons of these measurements with model predictions may also provide new sensitivity to α-clustering in the oxygen nucleus [88]. Compared to heavy-ion collisions, the v_2 in light-ions shows a much weaker centrality dependence, underscoring the enhanced role of EbE geometry fluctuations in these systems. As a function of event multiplicity, the v_3 values in O+O and Ne+Ne are quantitatively similar to those measured in Pb+Pb, Xe+Xe, and p+Pb collisions.

These light-ion results offer stringent constraints on hydrodynamic models incorporating *ab initio* nuclear-structure inputs, and provide unique insight into how nucleon-level geometry influences collective flow in small QGP droplets. Future comparisons with detailed model calculations could further constrain key medium and geometric properties, including the temperature dependence of η/s, the role of the pre-hydrodynamic phase, mechanisms of energy deposition in hadronic collisions, and other initial-state effects.

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